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 Studierichting: Informatica
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Bereken de volgende limiet:

$$\begin{aligned}
 & \lim_{x \rightarrow \pi} \frac{\tan 3x \left(\sin 2x - 3 \cos \frac{x}{2} \right)}{2 \cos^2 x + \cos x - 1} \\
 & \Rightarrow x = \pi + h, \quad h \rightarrow 0 \\
 & = \lim_{h \rightarrow 0} \frac{\tan(3\pi + 3h) \left(\sin(2\pi + 2h) - 3 \cos \left(\frac{\pi}{2} + \frac{h}{2} \right) \right)}{2 \cos^2(\pi + h) + \cos(\pi + h) - 1} \\
 & = \lim_{h \rightarrow 0} \frac{\tan(3h) \left(\sin(2h) - 3 \left(-\sin \frac{h}{2} \right) \right)}{2(-\cos h)^2 + (-\cos h) - 1} \\
 & = \lim_{h \rightarrow 0} \frac{\tan(3h) \left(\sin(2h) + 3 \sin \frac{h}{2} \right)}{2 \cos^2 h - \cos h - 1} \\
 & = \lim_{h \rightarrow 0} \frac{\frac{\tan(3h)}{3h} \cdot 3h \left(\frac{\sin(2h)}{2h} \cdot 2h + 3 \frac{\sin \frac{h}{2}}{\frac{h}{2}} \cdot \frac{h}{2} \right)}{2 \cos^2 h - \cos h - 1} \\
 & = \lim_{h \rightarrow 0} \frac{1 \cdot 3h \left(1 \cdot 2h + 3 \cdot 1 \cdot \frac{h}{2} \right)}{2(1 - h^2) - \left(1 - \frac{h^2}{2} \right) - 1} \\
 & = \lim_{h \rightarrow 0} \frac{3h \left(2h + \frac{3h}{2} \right)}{-2h^2 + \frac{h^2}{2}} \\
 & = \lim_{h \rightarrow 0} \frac{3h \cdot \frac{7h}{2}}{\frac{3h^2}{2} - \frac{2}{2}} \\
 & = \lim_{h \rightarrow 0} \frac{21h^2}{3h^2 - 2} \\
 & = -7
 \end{aligned}$$

References